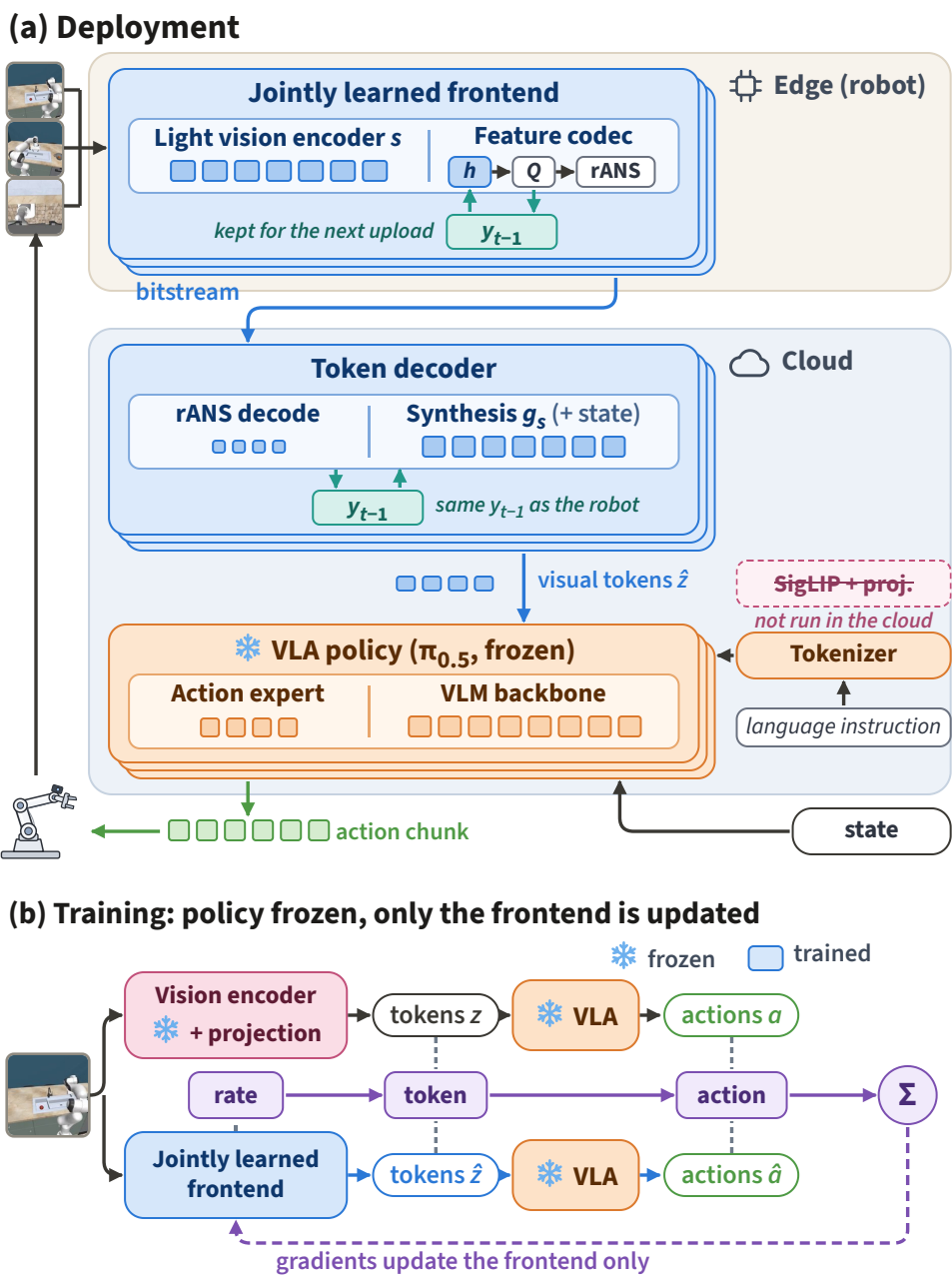

(a) Deployment
Edge (robot)
Jointly learned frontend
Light vision encoder s
Feature codec
h
Q
rANS
kept for the next upload
y_{t-1}
bitstream
Cloud
Token decoder
rANS decode
Synthesis g_s (+ state)
y_{t-1}
same y_{t-1} as the robot
visual tokens $\hat{z}$
SigLIP + proj.
not run in the cloud
VLA policy ($\pi_{0.5}$, frozen)
Action expert
VLM backbone
Tokenizer
language instruction
action chunk
state
(b) Training: policy frozen, only the frontend is updated
frozen
trained
Vision encoder + projection
tokens z
VLA
actions a
rate
token
action
Σ
Jointly learned frontend
tokens $\hat{z}$
VLA
actions $\hat{a}$
gradients update the frontend only